

Om Kumar

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EDUCATION

Manipal Institute of Technology

B.Tech in Mathematics and Computing

Manipal, Karnataka

July 2023 – Present

- CGPA: 8.47 / 10.0
- Relevant Coursework: Statistics, Probability Theory, Real Analysis, Linear Algebra, Optimization, Machine Learning

EXPERIENCE

Netweb Summer Research Intern

NISER Bhubaneswar (Hybrid)

May 2025 – Present

- Developed a post hoc floater removal pipeline for 3DGS that eliminates spurious Gaussian primitives without retraining; reduces model size to **20.4%** of the original Gaussian count while improving reconstruction quality by **+4.75 dB PSNR**
- Developed a multi-view diffusion framework for enhancing novel view synthesis in sparse, under-constrained regions, achieving up to **+10% PSNR**, **+15% SSIM**, and **–20% LPIPS** over baselines
- Improved 3D reconstruction pipeline for heritage sites, processing **TB-scale** multi-view image datasets and accelerating rendering by almost **1000×** using optimized point clouds, Gaussian splats, and mesh representations

Research Intern

Kharagpur Learning, Imaging and Visualization Group, IIT Kharagpur (Remote)

Nov 2024 – Present

- Co-developed an **arbitrary-scale super-resolution** model for ultrasonography, outperforming bicubic and baseline CNNs on PSNR/SSIM
- Built a fully reproducible, production-ready training pipeline with automated data versioning and experiment tracking, enabling large-scale experiments across **100K+** medical images
- Engineered automated dataset acquisition and preprocessing workflows on distributed GPU clusters, reducing manual setup time by **70%**
- Currently developing a multi-task, multi-modal foundation model for medical imaging with cross-dataset harmonization

Research Assistant

Manipal Institute of Technology

Mar 2024 – Sept 2024

Manipal, Karnataka

- Applied Generative Flow Networks (Bengio et al.) to generate novel drug candidates for *S. Aureus*, expanding candidate space beyond rule-based baselines

AI Subsystem Team Member

Project MANAS

Mar 2024 – Present

Manipal, Karnataka

- Achieved **3rd place globally** at IGVC 2025 (USA) in the **AutoNav Driverless Challenge**, competing as a fully autonomous ground vehicle
- Developed and deployed real-time **ODLC (Object Detection, Localization & Classification)** pipeline achieving **sub-30 ms** inference on embedded hardware using ONNX-optimized models
- Built **end-to-end autonomous navigation stack** including perception, path planning, and control for driverless operation

- Designed **efficient probabilistic occupancy grid mapping** framework, reducing memory overhead by **40%** for real-time decision-making
- Won **1st place** among autonomous drones and **3rd overall** (17 teams) at ISDC 2025; built monocular depth-based 3D terrain mapping

Part-Time Teacher

2019 – Present

Gurukul Classes (Remote)

- Delivering 1-on-1 mathematics and computer science to international students with **100%** client retention

LEADERSHIP AND ACTIVITIES

Team Manager

August 2025 – Present

Project MANAS

Manipal, Karnataka

- Heading Research, Marketing, Outreach, Budgeting, and General Operations for **Project MANAS** – The official AI and Robotics Research team of MIT Manipal

Organizing Committee

October 2024

TechTatva – MIT Manipal's Annual Technical Fest

Manipal, Karnataka

- Helped organize a paper presentation event on behalf of Research Society Manipal.

Associate

Oct. 2023 – Sept. 2024

Entrepreneurship Cell, MIT Manipal

Manipal, Karnataka

- Organized Manipal Entrepreneurship Summit 2024; designed reports and newsletters.

Local Committee Member

Oct. 2023 – Sept. 2024

IAESTE (International Association for Exchange of Students for Technical Experience)

Manipal, Karnataka

- Hosted international interns and produced content for conferences.

PROJECTS

Open Source Contributor - Project Tirtha | Python, PyTorch, Nerfstudio

Github

Integrated VGGT-based reconstruction, achieving $\sim 1000\times$ faster inference compared to baseline pipelines; served as the deployment backbone for the NISER heritage-site research project

Gurukul | Node.js, Next.js, React, Supabase, Vercel

Github

Built a full-stack tutoring platform, enabling online onboarding and increasing service reach beyond local geography

Monocular Depth Estimation for Terrain Mapping | PyTorch, OpenCV

Github

Generated depth from UAV RGB imagery and integrated it with path planning for autonomous exploration (ISDC 2025)

Object Detection and Localization | PyTorch, ROS2

Github

Trained and deployed ODLC models on an autonomous ground robot, enabling real-time perception during IGVC 2025

Automated Tender Acquisition (GeM) | Python, Selenium

Github

Automated government tender submissions, reducing manual effort from hours to minutes per tender

Ultrasound Image Super-Resolution | PyTorch

Github

Developed a lightweight attention-based U-Net improving diagnostic clarity over standard interpolation methods

SKILLS

Technical: Python, C++, Java, PyTorch, ROS/ROS2, Docker, Git, Gazebo, MuJoCo

Languages: English (B2), Spanish (A1)

Interests: Basketball, Writing, Swimming